



Compliance & Standards Matrix Bio-Sign™ Technical Whitepaper

Marrow-Locked Biometric Authentication Explained

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Executive Summary

Bio-Sign™ is REALAIID's proprietary **marrow-locked biometric authentication** protocol. It provides cryptographically verifiable **genuine presence** by anchoring identity to deep physiological signals that are extremely difficult to spoof. The system is designed for full compliance with NIST SP 800-63, the NO FAKES Act, and MARSEC maritime security standards while maintaining strict privacy protections.

1. Compliance & Standards Matrix

Standard / Regulation	REALAIID Bio-Sign™ Alignment	Compliance Level
NIST SP 800-63-4	Supports IAL3 / AAL3 through multi-factor liveness detection and cryptographic proofs	High
NO FAKES Act (2026)	Provides technical infrastructure for likeness protection and digital replica rights enforcement	Full
MARSEC / MTSA	Enables secure, biometric embarkation and port access control with forensic audit trails	High
BIPA / GDPR / CCPA	Zero raw biometric data stored or transmitted – stylized hashing only	Full
ISO 18013-5 (mDL)	Supports mobile driver's license inheritance and sovereign credential bridging	Strong

2. Bio-Sign™ Technical Architecture

Marrow-Locked Authentication



Bio-Sign™ goes beyond surface-level biometrics by measuring physiological signals from deeper tissue layers (vascular patterns, thermal resonance, and subtle frequency signatures inside the hand or palm). These signals are inherently tied to a living human body in real time.

Core Technical Features:

- **Stylized Hashing:** Only non-reversible cryptographic proofs are generated on-device.
- **Zero Raw Data Exposure:** No biometric templates are stored centrally or transmitted.
- **Real-Time Liveness:** Detects synthetic or replay attacks with high accuracy.
- **Handshake Knot™ Integration:** Combines with the full 13-Bridge Mainframe for sovereign identity.
- **Forensic Logging:** Every verification is cryptographically signed and timestamped for audit.

How It Works:

1. User presents hand to reader or uses mobile camera.
2. Device captures marrow-level signals locally.
3. Cryptographic proof is generated on-device.
4. Proof is sent to verifier (gate, terminal, wallet, etc.).
5. REALAIID backbone validates against the user's sovereign root.

3. Security & Privacy Guarantees

- **Privacy-First Design:** Compliant with BIPA, GDPR, and CCPA by default.
- **Anti-Spoofing:** Resistant to photos, videos, masks, and deepfakes.
- **Duress Mode:** Optional subtle signals for emergency situations.
- **Key Rotation:** Rotate-on-demand infrastructure.

4. Real-World Applications

- **EmbarkSecure™:** Contactless cruise ship and port boarding.
- **SentryBoard™:** Terminal and facility access control.
- **Bio-Pay™:** High-security financial transactions.
- **Supreme Ruler Tier:** Full sovereign identity management.

Conclusion & Next Steps



Bio-Sign™ represents the evolution of digital identity from static credentials to dynamic, biologically anchored truth. It provides the technical foundation for sovereign identity in the age of AI-generated content.

Call to Action

We welcome technical reviews, pilot programs, and partnership discussions with government agencies, maritime operators, and technology providers.

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